

# Charlie Marx

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## EDUCATION

**Stanford University**  
Ph.D. in Computer Science  
Advisor: Stefano Ermon

2020 - PRESENT

**Haverford College**  
B.S. in Computer Science, *highest honors*  
B.S. in Mathematics, *highest honors*  
GPA: 3.96/4.0, *magna cum laude*

2016 - 2020

## AWARDS & HONORS

**NSF Graduate Research Fellowship**  
**Stanford School of Engineering Fellowship**  
**Phi Beta Kappa**  
**CRA Outstanding Undergraduate Researcher Runner-Up**  
**Arnold and Mabel Beckman Foundation Award**  
**Farmers Insurance Standout Student of the Year**

2021  
2020  
2020  
2019  
2018  
2016

## SELECTED WORK

[Calibrated Probabilistic Forecasting for Arbitrary Sequences](#)  
**Charles Marx**, Volodymyr Kuleshov, Stefano Ermon  
TMLR – *Transactions on Machine Learning Research*, 2025

[Calibration by Distribution Matching: Trainable Kernel Calibration Metrics](#)  
**Charles Marx\***, Sofian Zalouk\*, Stefano Ermon  
NeurIPS – *Conference on Neural Information Processing Systems*, 2023

[Predictive Multiplicity in Classification](#)  
**Charles Marx**, Flavio du Pin Calmon, Berk Ustun  
ICML – *International Conference on Machine Learning*, 2020  
Previously Selected for Oral Presentation at 2019 NeurIPS HCML Workshop

## PUBLICATIONS

1. [Online Calibrated and Conformal Prediction Improves Bayesian Optimization](#)  
Shachi Deshpande, **Charles Marx**, Volodymyr Kuleshov  
AISTATS – *Artificial Intelligence and Statistics*, 2024
2. [Calibration by Distribution Matching: Trainable Kernel Calibration Metrics](#)  
**Charles Marx\***, Sofian Zalouk\*, Stefano Ermon  
NeurIPS – *Conference on Neural Information Processing Systems*, 2023
3. [But Are You Sure? An Uncertainty-Aware Perspective on Explainable AI](#)  
**Charles Marx**, Youngsuk Park, Hilaf Hasson, Yuyang Wang, Stefano Ermon, Jun Huan  
AISTATS – *Artificial Intelligence and Statistics*, 2023
4. [Classification with Conceptual Safeguards](#)  
Hailey Joren, **Charles Marx**, Berk Ustun  
ICLR – *International Conference on Learning Representations*, 2023
5. [Modular Conformal Calibration](#)  
**Charles Marx\***, Shengjia Zhao\*, Willie Neiswanger, Stefano Ermon  
ICML – *International Conference on Machine Learning*, 2022
6. [Predictive Multiplicity in Classification](#)  
**Charles Marx**, Flavio du Pin Calmon, Berk Ustun  
ICML – *International Conference on Machine Learning*, 2020  
Previously Selected for Oral Presentation at 2019 NeurIPS HCML Workshop
7. [Disentangling Influence: Using Disentangled Representations to Audit Model Predictions](#)  
**Charles Marx**, Richard Phillips, Sorelle Friedler, Carlos Scheidegger, Suresh Venkatasubramanian  
NeurIPS – *Conference on Neural Information Processing Systems*, 2019

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|                                      | <p>8. <a href="#">Examining Inputs of Biogenic and Oil-Derived Hydrocarbons in Surface Waters Following the Deepwater Horizon Oil Spill</a><br/>Helen White, <b>Charles Marx</b>, David Valentine, Charles Sharpless, Christoph Aeppli, Kelsey Gosselin, Veronika Kivenson, Rachel Liu, Robert Nelson, Sean Sylva, Christopher Reddy<br/><i>ACS Earth and Space Chemistry</i>, 2019</p> <p>9. <a href="#">Rapid Identification of Deepwater Horizon Oil Residues Using X-Ray Fluorescence</a><br/>Anna Michel, Alexandra E. Morrison, <b>Charles Marx</b>, Helen White<br/><i>ACS Environmental Science &amp; Technology Letters</i>, 2018</p>                                                                                                                                                                                                                                                              |  |
| RESEARCH &<br>INDUSTRY<br>EXPERIENCE | <p><b>Amazon (Language Modeling)</b> <i>Applied Science Intern</i> SUMMER 2023<br/>Designed algorithms for detecting hallucinations in large language models.<br/><i>Advisors:</i> Wooseok Ha and Christian Bock</p> <p><b>Amazon (Time Series ML)</b> <i>Applied Science Intern</i> SUMMER 2022<br/>Designed machine learning methods for quantifying uncertainty in model explanations.<br/><i>Advisors:</i> Youngsuk Park and Hilaf Hasson</p> <p><b>International Food Policy Research Institute</b> <i>Intern</i> SUMMER 2020<br/>Built computer vision systems to classify land usage in Africa using satellite images.<br/><i>Advisors:</i> Channing Arndt and Zhe Guo</p> <p><b>Harvard University</b> <i>Research Assistant</i> SUMMER 2019<br/>Developed integer programming tools to measure underspecification in machine learning tasks.<br/><i>Advisors:</i> Flavio Calmon and Berk Ustun</p> |  |
| TEACHING                             | <p><b>Stanford CS 228: Probabilistic Graphical Models (Head TA)</b> 2024</p> <p><b>Stanford CS 229: Machine Learning</b> 2024</p> <p><b>Haverford CMSC 360: Machine Learning</b> 2020</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
| INVITED &<br>CONTRIBUTED<br>TALKS    | <p><b>Data for Sustainability Conference</b> <i>Stanford, CA</i> 2023<br/>Probabilistic Forecasting for Science<br/><i>Best Talk Award</i></p> <p><b>ICML Conference</b> <i>Baltimore, MD</i> 2022<br/>Modular Conformal Calibration</p> <p><b>Simons Institute for the Theory of Computing</b> <i>Berkeley, CA</i> 2019<br/>Disentangling Influence</p> <p><b>INFORMS Annual Meeting</b> <i>Seattle, WA</i> 2019<br/>Predictive Multiplicity in Classification</p> <p><b>FAT* Conference</b> <i>New York, NY</i> 2018<br/>Auditing Black Box Models (60-minute tutorial)<br/><i>with Carlos Scheidegger and Suresh Venkatasubramanian</i></p> <p><b>GOMOSSES Conference</b> <i>New Orleans, LA</i> 2018<br/>Examining Inputs of Biogenic and Oil-Derived Hydrocarbons</p>                                                                                                                                  |  |
| ACADEMIC<br>SERVICE                  | <p><b>Reviewing:</b> NeurIPS ('22, '24), ICML ('22), KDD ('20), IEEE JSAT ('23), ACM JRC ('23)<br/><b>Program Committee:</b> FAccT (2022, 2024), NeurIPS Workshop on Fair AI in Finance (2020)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
| EXTRA—<br>CURRICULAR<br>ACTIVITIES   | <p><b>Captain of Haverford College Men's Club Soccer</b> 2018 – 2019</p> <p><b>Leader of Haverford College Outdoors Club</b> 2017 – 2018</p> <p><b>President of Furniture for Newly Housed Families Organization</b> 2015 – 2017</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
| SKILLS &<br>INTERESTS                | <p><b>Languages:</b> <i>Fluent:</i> Python, R, Bash · <i>Familiar:</i> SQL, C</p> <p><b>Tools:</b> PyTorch, Tensorflow, Pandas, CPLEX</p> <p><b>Interests:</b> Backpacking, Basketball, Cooking, Soccer</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |